

Exploring RGB and Video V-JEPA 2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

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Abstract. Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. Camera-based perception is attractive for these settings given its light weight, low cost, availability, and compatibility with UAV size, weight, and power constraints. This motivates methods that can infer geometry, semantics, and free space from visual observations, making Monocular Depth Estimation (MDE) and Semantic Scene Completion (SSC) key components of UAV 3D perception that enable 3D scene understanding. Aerial imagery introduces altitude-dependent scale variation, oblique viewpoints, weak texture, and different occlusion patterns, demanding domain-specific solutions. Using OccuFly, a real-world aerial dataset with RGB images, metric depth maps, and semantic occupancy grids, we study whether V-JEPA 2.1 foundation features transfer to aerial MDE and SSC. Our ablations progress from V-JEPA 2.1 image-spatial & video-spatiotemporal feature handling, from linear probing to DPT (Dense Prediction Transformer)-style multiscale decoding, FiLM (Feature-wise Linear Modulation)-based altitude & intrinsics conditioning for MDE, followed by injecting conditioned depth-estimation priors, deploying a learnable UAV pose-aware multi-view stereo depth-estimation module, and transferring spatiotemporal features for SSC. Through this staged analysis, we separate the contributions of representation transfer, altitude conditioning, and depth-estimation variants to improving aerial SSC.

Keywords: Aerial · 3D Scene Understanding · UAV Perception · Monocular Depth Estimation · Semantic Scene Completion · Semantic Occupancy Prediction · V-JEPA 2.1 · Video Foundation models · 2D-3D lifting · Pose-aware multi-view stereo depth · OccuFly

1 Introduction

Aerial 3D scene understanding is central to autonomous uncrewed aerial vehicle (UAV) operation in cargo delivery, search and rescue, inspection, and human-UAV collaboration. In these settings, a perception system must infer not only visible image semantics, but also metric depth, free space, and the semantic occupancy of partially observed volumes. RGB cameras are attractive for UAVs because they are lightweight, inexpensive, and compatible with strict size, weight,

and power constraints. However, monocular 3D perception remains difficult: image formation collapses scale, depth, and occlusion into a two-dimensional signal, and aerial views introduce altitude-dependent scale changes, oblique/top-down viewpoints, weak texture, and occlusion patterns that differ from ground-level driving scenes.

Most progress in semantic scene completion (SSC) and camera-based 3D perception has been driven by indoor and autonomous-driving benchmarks. SSCNet introduced semantic completion from depth input [24]; MonoScene extended the task to monocular RGB [5]; and subsequent methods such as VoxFormer [12], TPVFormer [10], SurroundOcc [26], OpenOccupancy [25], CGFormer [31], and FoundationSSC [6] improved view transformation, voxel attention, and semantic-geometric fusion. These designs demonstrate the value of explicit 3D representations.

OccuFly [8] enables a focused study of this transfer problem. It provides aerial RGB images, metric depth maps, camera calibration, poses, altitude metadata, and semantic occupancy grids across multiple scenes and flight heights. Unlike 2D UAV segmentation datasets such as UAVid [15] or reconstruction-oriented resources such as UrbanScene3D [14], OccuFly supports joint evaluation of monocular depth estimation (MDE), multiview geometry, and SSC from real aerial observations. We use it to ask whether modern foundation representations provide transferable geometric and semantic priors for UAV 3D perception, and whether calibrated neighboring views improve SSC beyond monocular depth priors.

We investigate frozen V-JEPA 2.1 [17] image and video features for aerial MDE and depth-aware SSC. V-JEPA-style models learn by predicting latent representations rather than reconstructing pixels [2, 3]; V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision, and unified image/video tokenization. Rather than finetuning the backbone, we keep it frozen and vary only the decoder, metadata conditioning, temporal adapter, and lifting mechanism. This protocol makes the study diagnostic: improvements can be attributed to the downstream structure used to extract depth and occupancy from the representation.

Our evaluation has three parts. First, we compare linear probes, DPT-style multiscale decoders [21], Simple Feature Pyramid (SFP) adapters inspired by ViTDet [11], and Feature-wise Linear Modulation (FiLM) [19] using altitude and camera intrinsics. This separates linear depth accessibility, dense reconstruction capacity, final-layer pyramid adaptation, and metric scale conditioning. Second, we compare RGB features with spatio-temporal V-JEPA 2.1 video features using a dense temporal-attention adapter that converts tubelet tokens into target-frame maps. Third, we evaluate SSC variants inspired by CGFormer and FoundationSSC, replacing or augmenting their geometry pathways with V-JEPA-derived depth probabilities and a pose-aware multiview stereo (MVS) cost-volume branch based on calibrated UAV views.

Our contributions are:

- A systematic frozen-backbone evaluation of V-JEPA 2.1 RGB and video representations for aerial metric depth estimation on OccuFly.

- A controlled comparison of linear probes, DPT-style decoders, SFP-adapted final-layer decoders, and altitude/intrinsics-conditioned FiLM variants.
- An analysis of spatio-temporal feature transfer through dense temporal attention over V-JEPA 2.1 video tokens.
- A depth-to-SSC study that injects V-JEPA-derived depth maps and depth distributions into CGFormer- and FoundationSSC-inspired aerial SSC variants.
- A pose-aware FoundationSSC variant that replaces monocular depth lifting with calibrated multiview cost-volume depth estimation from neighboring UAV frames.

2 Related Work

2.1 Monocular depth estimation and dense prediction decoders

Monocular depth estimation is ambiguous because metric depth must be inferred from perspective, texture, object scale, layout, and learned priors. Early deep methods used coarse-to-fine CNNs and scale-aware objectives [7]. Later work improved transfer through stronger encoders, multiscale dense decoders, and broader supervision. MiDaS showed the benefit of mixing heterogeneous datasets [22]; DPT reassembled transformer tokens into multiscale image-like features for dense prediction [21]; ZoeDepth combined relative-depth generalization with metric specialization [4]; and Depth Anything/Depth Anything V2 scaled monocular depth training with large pseudo-labeled and synthetic data [27, 28]. Aerial depth remains a distinct transfer problem because altitude changes scale, many images lack ground-level horizon cues, repetitive roofs and vegetation reduce texture, and far-range depth errors can degrade voxel lifting. We therefore compare frozen linear probes, DPT-style decoders, SFP adaptation inspired by ViTDet [11], and altitude/intrinsics FiLM conditioning [19] on the same V-JEPA 2.1 features.

2.2 Multiview stereo and cost-volume geometry

Multiview stereo (MVS) estimates depth by matching calibrated observations across views. MVSNet constructs plane-sweep cost volumes, regularizes them with 3D convolutions, and infers depth from a distribution over depth hypotheses [29]. Recurrent MVSNet reduces memory by processing hypotheses recurrently [30]. This projective matching principle is directly relevant to UAV SSC: when poses are available, target/source correspondence can provide geometric evidence complementary to monocular priors. Our Pose-MVS variants adapt this idea to frozen V-JEPA 2.1 video features by predicting a categorical depth distribution aligned with FoundationSSC lifting bins, then using that distribution to place semantic context into the OccuFly voxel grid.

2.3 Self-supervised visual foundation models

Self-supervised visual foundation models learn transferable representations without task-specific labels. Masked autoencoders scale ViTs through high-ratio masked image reconstruction [9]. I-JEPA predicts target-block representations from context representations in latent space [1]. DINOv2 produces general-purpose visual features from large-scale self-supervision [18], and DINOv3 emphasizes high-resolution dense features and transfer across natural and aerial imagery [23]. Video representation learning extends these ideas to temporal structure: V-JEPA predicts masked spatio-temporal features in latent space [3], V-JEPA 2 scales the approach to large image/video data [2], and V-JEPA 2.1 modifies the recipe toward dense image/video features [17]. We evaluate V-JEPA 2.1 through downstream depth, cost-volume, and SSC probes rather than through representation visualization or global recognition metrics.

2.4 2D-to-3D lifting and camera-based semantic scene completion

Semantic scene completion jointly predicts free/occupied voxels and occupied semantic labels. SSCNet introduced the task from single-view depth [24], and MonoScene lifted monocular RGB features into 3D [5]. Outdoor camera-based SSC then moved toward sparse voxel queries, deformable attention, and explicit view transformation. VoxFormer initializes visible voxel queries using estimated depth [12]; TPVFormer represents 3D space with three orthogonal planes [10]; and SurroundOcc/OpenOccupancy extend occupancy prediction to surround-view driving settings [25, 26].

Depth-aware lifting is a recurring theme. Lift-Splat-Shoot predicts a ray-wise depth distribution and splats image features into a shared spatial grid [20]; BEVDepth shows that explicit depth supervision improves camera-based BEV perception [13]. CGFormer addresses depth ambiguity with context-dependent voxel queries and 3D pixel-space deformable cross-attention [31]. FoundationSSC decouples semantic and geometric priors at both source and pathway levels before hybrid 3D fusion [6]. Our SSC experiments follow this line but focus on aerial observations: V-JEPA 2.1 features provide semantic context, while monocular depth, video aggregation, and pose-aware MVS provide alternative geometry priors under a shared OccuFly protocol.

3 Method

We evaluate frozen V-JEPA 2.1 image and video features for two aerial 3D tasks: metric depth estimation and semantic scene completion (SSC). The framework is organized as a controlled sequence of probes and decoders. For MDE, we compare linear probing, DPT-style dense decoding, final-layer SFP adaptation, FiLM metadata conditioning, and video-token aggregation. For SSC, we use CGFormer- and FoundationSSC-inspired designs because their image-context and geometry pathways can be varied independently. This lets us isolate the contributions of frozen representation transfer, dense decoding, altitude/intrinsics

RGB		V-JEPA 2.1 ViTG		Video	
				Dense Temporal Attention (DTA)	
MDE - Depth Heads		SSC		MDE - Depth Heads	
				SSC	
1. Linear Probe RGB	CGFormer variant		2. Linear Probe Video	6.FSSC_video_context DTA_DPT_depth_DTA_FILM_DPT	
	Image Context: Direct			Image Context: DTA_DPT	
3. Simple Feature Pyramid - Dense Prediction Head (SFP_FILM_DPT)	Geometry:	Depth priors		Geometry: DTA_FILM_DPT	
		1. DAV2 Finetuned Occufly			
4. Multi Layer - Dense Prediction Head (DPT)		2. FILM_DPT	4.FSSC_video_context DTA_DPT_depth_DTA_mvs		
	3.FSSC_RGB_context DPT_depth_Film_DPT		Image Context: DTA_DPT		
5. Multi Layer - Featurewise Linear Modulation - Dense Prediction Head (FILM_DPT_RGB)	Image Context: DPT		6. DTA_FILM_DPT_Video	Geometry: DTA_Pose_mvs	
	Geometry: FILM_DPT				
	4.FSSC_RGB_context FILM_DPT_depth_FILM_DPT				5.FSSC_video_context DTA_DPT_depth_DTA_FILM_mvs
	Image Context: FILM_DPT		Image Context:DTA_FILM_DPT		
		Geometry: FILM_DPT	Geometry: FILM_Pose_MVS		

Fig. 1: Evaluation overview for the dense feature-transfer study. The figure summarizes the progression from frozen V-JEPA 2.1 RGB/video representations to monocular and multiview depth estimation heads, followed by the SSC variants that use the selected context and geometry pathways.

conditioning, temporal aggregation, monocular depth priors, and pose-aware multiview geometry. Fig. 1 summarizes the evaluation flow.

3.1 Problem Formulation and Notation

Each OccuFly sample is represented as

$$s_t = (I_t, D_t, M_t, Y_t, K_t, h_t, T_t^{w \rightarrow c}), \quad (1)$$

where $I_t \in \mathbb{R}^{3 \times H \times W}$ is an RGB frame, D_t is metric depth, $M_t \in \{0, 1\}^{H \times W}$ is the valid-depth mask, Y_t is the semantic occupancy grid, K_t is the camera intrinsic matrix, h_t is UAV altitude, and $T_t^{w \rightarrow c} \in \mathbb{R}^{4 \times 4}$ is the world-to-camera pose.

The monocular depth, pose-aware multiview depth, and SSC mappings are

$$\begin{aligned} f_D^{\theta_D} &: (I_t, K_t, h_t) \mapsto \hat{D}_t, \\ f_{D, \text{MVS}}^{\theta_M} &: (\mathcal{V}_t, h_t) \mapsto (\hat{D}_t^{\text{MVS}}, P_t^{D, \text{MVS}}), \\ f_S^{\theta_S} &: (\Phi_t, K_t, h_t, \hat{D}_t, P_t^D) \mapsto \hat{Y}_t. \end{aligned} \quad (2)$$

The multiview input is

$$\mathcal{V}_t = \{(I_{t_i}, K_{t_i}, T_{t_i}^{w \rightarrow c})\}_{i=0}^{N_s}, \quad t_0 = t, \quad (3)$$

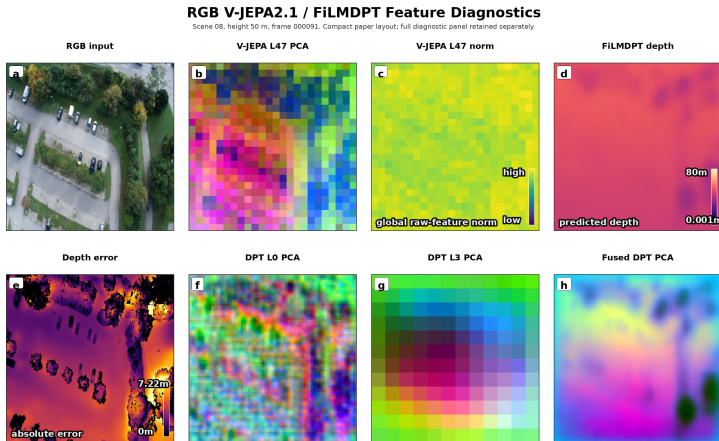


Fig. 2: Compact RGB V-JEPA 2.1/FiLM-DPT feature diagnostic. PCA views of frozen and decoder-processed features illustrate spatially organized cues used by downstream dense prediction.

where t_0 is the target view and $t_1, \dots, t_{N_s}$ are calibrated source views. We use two views in this study. The V-JEPA 2.1 encoder is frozen in all experiments; trainable parameters are restricted to the depth heads θ_D , MVS head θ_M , and SSC modules θ_S according to the evaluated variant. Φ_t denotes the target visual context, $\hat{D}_t$ the dense metric depth estimate, P_t^D the ray-wise depth probability tensor used for lifting, and $\hat{Y}_t$ the predicted semantic occupancy grid.

3.2 Frozen V-JEPA 2.1 Feature Extraction

Let E_θ be the frozen V-JEPA 2.1 encoder [17]. For the ViT-Gigantic checkpoint used here, the input resolution is 384×384 , patch size is 16, the spatial grid is 24×24 , the feature dimension is $C = 1664$, and the 48 transformer blocks are indexed from 0 to 47.

RGB mode. For a target frame I_t , we use either the final feature map or four intermediate layers,

$$\mathcal{L}_G = \{11, 23, 37, 47\}, \quad F_t^{(l)} = E_\theta^{(l)}(I_t) \in \mathbb{R}^{1664 \times 24 \times 24}, \quad l \in \mathcal{L}_G. \quad (4)$$

The linear probe uses only $F_t^{(47)}$. The DPT variants consume selected intermediate maps and reassemble them into multiscale dense features following DPT [21].

Video mode. We form a four-frame target-centered clip

$$X_t = [I_{t+\tau_1}, I_{t+\tau_2}, I_{t+\tau_3}, I_{t+\tau_4}] \in \mathbb{R}^{3 \times 4 \times 384 \times 384}, \quad (5)$$

$$[\tau_1, \tau_2, \tau_3, \tau_4] = [-1, 0, 0, +1].$$

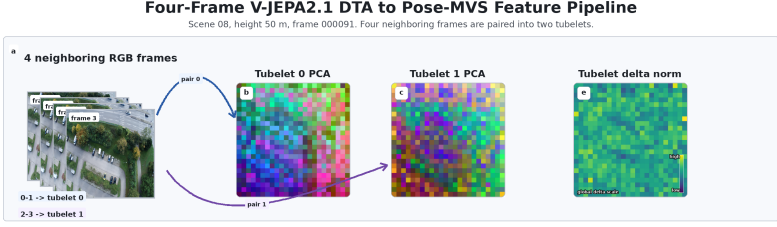


Fig. 3: Compact video-token diagnostic for the four-frame V-JEPA 2.1 pathway. Consecutive RGB frames form two tubelet positions whose PCA projections and feature-delta norm summarize the temporal cues exposed to DTA and pose-aware MVS modules.

The duplicated zero offset keeps the clip target-centered while matching the V-JEPA 2.1 tubelet size. Because the OccuFly preprocess files used here do not include capture timestamps, temporal neighborhoods are defined by frame offset rather than physical time.

For tubelet size 2, the four-frame clip produces $T' = 2$ temporal tubelet positions:

$$G_t^{(l)} = E_\theta^{(l)}(X_t) \in \mathbb{R}^{1664 \times T' \times 24 \times 24}. \quad (6)$$

We test two adapters. The video linear probe folds the tubelet axis into channels,

$$G_t^{(47)} \in \mathbb{R}^{1664 \times T' \times 24 \times 24} \mapsto \tilde{F}_t \in \mathbb{R}^{(1664T') \times 24 \times 24}, \quad (7)$$

$$\tilde{F}_t[t'C + c, u, v] = G_t^{(47)}[c, t', u, v].$$

For $T = 4$, $\tilde{F}_t \in \mathbb{R}^{3328 \times 24 \times 24}$.

The Dense Temporal Attention (DTA) adapter converts video tokens into target-frame maps. For each $l \in \{11, 23, 37, 47\}$ and spatial location (u, v) , temporal descriptors are

$$S_{t,u,v}^{(l)} = [G_t^{(l)}[:, 1, u, v], \dots, G_t^{(l)}[:, T', u, v]] \in \mathbb{R}^{T' \times 1664}. \quad (8)$$

A lightweight learned-query attention block returns one feature vector per spatial cell,

$$\bar{F}_t^{(l)}[:, u, v] = \mu_t^{(l)}[:, u, v] + \alpha_l \left(\text{DTA}_l(S_{t,u,v}^{(l)}) - \mu_t^{(l)}[:, u, v] \right), \quad (9)$$

where $\mu_t^{(l)} = \text{mean}_\tau G_t^{(l)}[:, \tau, :, :]$ and the residual gate α_l is initialized near mean pooling. The maps $\{\bar{F}_t^{(l)}\}$ then have the same $1664 \times 24 \times 24$ form as RGB maps and can be passed to the DPT, feature-to-depth, or pose-aware MVS modules.

Input split	OccuFly train/validation/test samples
↓	image (I_t, K_t, h_t) , video clip (X_t, K_t, h_t) , or posed views $\mathcal{V}_t$
Frozen backbone	V-JEPA 2.1 features $F_t^{(l)}$ or $G_t^{(l)}$
↓	LP / SFP-DPT / DPT-FiLM / Video-FiLM-DPT / Pose-MVS-FiLM
Trainable head	depth-bin logits
↓	
Output	depth probabilities P_t^D and estimated depth $\hat{D}_t$ or MVS counterparts $(P_t^{D,MVS}, \hat{D}_t^{MVS})$

Fig. 4: Common depth-estimation flow. The V-JEPA 2.1 backbone is frozen, while the depth head is trained on the official OccuFly train/validation/test split.

3.3 Depth Estimation

All depth heads predict logits over $K_D = 256$ metric depth bins and convert them to a dense expectation,

$$p_{k,u,v} = \text{softmax}_k(z_{k,u,v}), \quad \hat{D}_{u,v} = \sum_{k=1}^{K_D} p_{k,u,v} d_k, \quad (10)$$

where d_k are bin centers. Monocular heads produce z from linear or DPT-style decoders; Pose-MVS produces z from a regularized projective cost volume.

Linear probes. The RGB and video linear probes test whether metric aerial depth is directly accessible from frozen final-layer features. The RGB probe uses $F_t^{(47)}$; the video probe uses $\tilde{F}_t$ from Eq. (7). Both use only a batch-normalized 1×1 prediction head, following dense frozen-feature probing practice [16, 23].

DPT Depth Decoders DPT-style decoders reassemble transformer maps into image-like features and fuse them with convolutional refinement [21]. For selected layers,

$$R_l = \rho_l \left(F_t^{(l)} \right), \quad R_l \in \mathbb{R}^{C_D \times H_l \times W_l}, \quad C_D = 256, \quad (11)$$

where ρ_l is the DPT reassembly/projection block. The maps are either fused directly or FiLM-conditioned before fusion.

SFP-DPT. The SFP variant tests whether the final V-JEPA 2.1 feature map can serve as a dense multiscale source without intermediate layers. Inspired by the Simple Feature Pyramid used with plain ViTs in ViTDet [11], it constructs

$$\text{SFP}(F_t^{(47)}) = \{S_4, S_2, S_1, S_{1/2}\}, \quad S_s \in \mathbb{R}^{256 \times (24s) \times (24s)}, \quad s \in \{4, 2, 1, 1/2\}. \quad (12)$$

The resulting pyramid is decoded by the same DPT-style depth head.

Table 1: Simplified component flow for the depth-head variants. Light-pink boxes denote the learnable components updated during training; unmarked V-JEPA 2.1 feature extraction remains frozen.

Variant	Data flow with representative shapes
RGB Probe	Linear $I_t \rightarrow F_t^{(47)}[1664, 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t[256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
Video Probe	Linear $X_t \rightarrow G_t^{(47)}[1664, T', 24, 24] \rightarrow \tilde{F}_t[(1664T'), 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t[256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
SFP-DPT	$I_t \rightarrow F_t^{(47)}[1664, 24, 24] \rightarrow \text{SFP}\{S_4, S_2, S_1, S_{1/2}\} \rightarrow \text{DPT} \rightarrow (\hat{D}_t, P_t^D)$, $S. \in \{[256, 96, 96], [256, 48, 48], [256, 24, 24], [256, 12, 12]\}$.
DPT	$I_t \rightarrow \{F_t^{(11)}, F_t^{(23)}, F_t^{(37)}, F_t^{(47)}\} \rightarrow \text{DPT reassembly/fusion} \rightarrow (\hat{D}_t, P_t^D)$.
FiLM-DPT	$I_t, K_t, h_t \rightarrow \{R_l\}, e_t \rightarrow \text{FiLM}(\{R_l\}, e_t) \rightarrow \text{DPT fusion} \rightarrow (\hat{D}_t, P_t^D)$.
Video DPT	FiLM- $X_t, K_t, h_t \rightarrow \{G_t^{(l)}\} \rightarrow \text{DTA} \rightarrow \{\bar{F}_t^{(l)}\}, e_t \rightarrow \text{FiLM-DPT} \rightarrow (\hat{D}_t, P_t^D)$.
Pose-MVS-FiLM	$\mathcal{V}_t, h_t \rightarrow G_t^{(l)} \rightarrow \text{DTA} \rightarrow \text{pose cost volume} \rightarrow \text{FiLM logits} \rightarrow (\hat{D}_t^{\text{MVS}}, P_t^{D, \text{MVS}})$.

FiLM-conditioned DPT. FiLM variants test whether altitude and camera calibration should modulate the frozen-feature decoder. The normalized metadata vector is

$$m_t = \left[\frac{h_t - 40}{10}, \frac{f_x}{W}, \frac{f_y}{H}, \frac{c_x}{W} - 0.5, \frac{c_y}{H} - 0.5 \right] \in \mathbb{R}^5, \quad (13)$$

which a small MLP maps to $e_t \in \mathbb{R}^{128}$. For each DPT level,

$$\tilde{R}_l = R_l \odot (1 + \gamma_l(e_t)) + \beta_l(e_t), \quad (14)$$

where γ_l and β_l are channel-wise affine projections broadcast spatially. They are initialized near zero so the decoder starts close to the unconditioned DPT path. For video FiLM-DPT, DTA first converts tubelet features to $\{\bar{F}_t^{(l)}\}$; the same FiLM-DPT decoder is then applied. Equivalently, at each spatial site,

$$\bar{F}_t^{(l)}(:, u, v) = \mu_t^{(l)}(:, u, v) + \alpha \left(\text{Expand}(\text{MLP}(\text{MHA}(q, K, V))) - \mu_t^{(l)}(:, u, v) \right), \quad (15)$$

with $\alpha = \sigma(a)$ initialized to approximately 0.1.

Pose-MVS-FiLM Pose-MVS-FiLM constructs a depth-indexed cost volume from calibrated target/source views, following learned MVS cost-volume match-

Table 2: Depth-estimation variants and the transfer question isolated by each design. Learnable components are shown in Table 1.

Variant	Purpose
Linear probe	Tests whether metric aerial depth is linearly accessible from frozen V-JEPA 2.1 features.
SFP-DPT	Tests whether the final V-JEPA 2.1 map can be converted into an effective dense pyramid without intermediate layers.
DPT	Tests whether selected intermediate layers improve dense metric depth over final-layer probing.
FiLM-DPT	Tests whether altitude/intrinsics conditioning improves RGB or video depth decoding.
Pose-MVS-FiLM	Tests whether calibrated multiview matching and meta-data conditioning improve the geometric prior.

ing [29]. For target pixel $\mathbf{u} = (u, v, 1)^\top$ and depth-bin center d_k ,

$$\mathbf{x}_{t,k,u,v} = d_k K_t^{-1} \mathbf{u}, \quad \mathbf{q}_{i,k,u,v} = \pi \left(K_i T_{t \rightarrow i} \begin{bmatrix} \mathbf{x}_{t,k,u,v} \\ 1 \end{bmatrix} \right), \quad (16)$$

where $\pi(\cdot)$ is perspective projection. Let A_t and A_i be target/source feature maps after video adaptation. Source features are sampled at projected coordinates and aggregated over source views,

$$\tilde{A}_{i,k,u,v} = \text{sample}(A_i, \mathbf{q}_{i,k,u,v}), \quad C_{t,k,u,v} = \text{Agg}_{i \in \mathcal{S}_t} \text{corr} \left(A_t(:, u, v), \tilde{A}_{i,k,u,v} \right). \quad (17)$$

A 3D CNN regularizes C_t and produces FiLM-conditioned depth logits,

$$P_t^{D,\text{MVS}} = \text{softmax}_k Z_t^{\text{MVS}}, \quad \hat{D}_{t,u,v}^{\text{MVS}} = \sum_{k=1}^{K_D} P_{t,k,u,v}^{D,\text{MVS}} d_k. \quad (18)$$

This branch uses the same 256 depth-bin centers and target-frame depth supervision as Eq. (10).

Altitude-Interpolation Depth Evaluation The standard OccuFly split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing, with altitudes 30, 40, and 50 m. We also evaluate altitude interpolation: training uses 30 and 50 m samples from scenes 01–08, validation uses 40 m samples from scenes 01–04, and testing uses 40 m samples from scenes 05–08,

$$\mathcal{D}_{\text{train}} = \{s_t : h_t \in \{30, 50\}, \text{scene}(t) \in \{01, \dots, 08\}\}, \quad (19)$$

$$\mathcal{D}_{\text{val}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{01, 02, 03, 04\}\}, \quad (20)$$

$$\mathcal{D}_{\text{test}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{05, 06, 07, 08\}\}. \quad (21)$$

This split tests whether FiLM conditioning supports metric scale interpolation rather than only fitting observed flight heights.

3.4 Semantic Scene Completion

The SSC benchmark separates image context from geometry. The context pathway provides semantic visual features; the geometry pathway provides monocular depth, depth probabilities, or pose-aware MVS priors. Both are lifted into a common 3D representation before fusion and semantic occupancy prediction. The goal is not to introduce a completely new SSC architecture, but to use architectures whose exposed pathways support controlled feature-transfer studies.

4 Experimental Setup and Results

4.1 Dataset and Tasks

We evaluate monocular depth estimation (MDE) and semantic scene completion (SSC) on OccuFly [8]. The standard split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing across altitudes of 30, 40, and 50 m. The depth split contains 14,804 training frames, 1,965 validation frames, and 3,842 test frames. RGB inputs are resized to 384×384 ; metric depth is clipped to $[0.001, 80]$ m and evaluated only over finite positive pixels that remain valid after mask-aware resizing. Video variants use consecutive frame-index neighborhoods because the released preprocess files provide frame indices and poses but no measured capture FPS.

For SSC, OccuFly labels are defined on a $192 \times 128 \times 128$ voxel grid in WHD order. The grid spans $x \in [-48, 48]$ m, $y \in [-32, 32]$ m, and $z \in [0, 64]$ m, with 0.5 m voxels. Unless stated otherwise, experiments are run on an NVIDIA DGX node with 8 NVIDIA H100 80GB GPUs.

4.2 MDE Benchmarking Results

Table 3 compares depth models on the standard OccuFly test split. Linear probing shows that frozen V-JEPA 2.1 features contain usable depth information, but

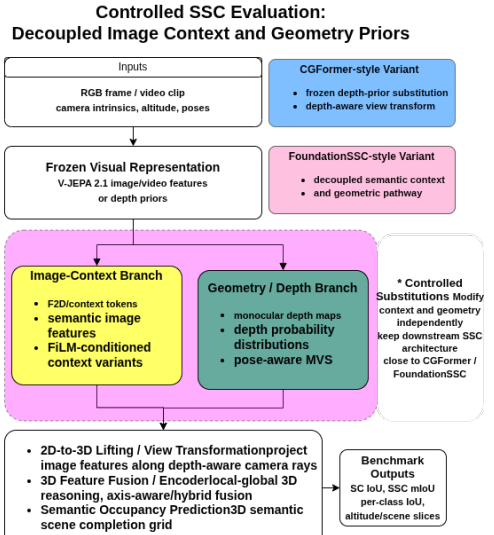


Fig. 5: Conceptual SSC flow with independently varied image-context and geometry-depth pathways.

Table 3: OccuFly test monocular depth estimation comparison. Published baseline values for MapAnythingV1.1, Metric3Dv2, Depth Anything 2, and Depth Anything 3 follow Table 5 of OccuFly [8]. Light pink marks the best RMSE.

Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
Linear Probe RGB	0.651	0.958	0.996	0.189	7.808	6.457	0.226
Linear Probe Video 4 frame clip	0.692	0.953	0.994	0.190	7.226	5.996	0.220
SFP_FiLM_DPT	0.808	0.953	0.993	0.143	5.160	3.732	0.189
DPT	0.469	0.795	0.898	0.267	11.471	9.227	0.411
FiLM_DPT_RGB	0.833	0.971	0.998	0.129	4.621	3.523	0.164
DTA_FiLM_DPT_Video	0.805	0.970	0.998	0.136	4.878	3.635	0.167
DepthAnything2_OccuFly [8]	0.834	0.976	0.999	0.134	4.844	4.193	0.112

dense decoding and metadata conditioning are much stronger. FiLM-DPT-RGB gives the best overall V-JEPA 2.1 result, with 4.621 m RMSE and 0.129 AbsRel, slightly improving RMSE over DepthAnything2-OccuFly while using the frozen V-JEPA 2.1 backbone.

Table 4 reports the altitude-interpolation split. The FiLM-DPT-RGB head remains accurate when trained at 30/50 m and tested at 40 m, supporting the use of altitude and intrinsics as scale-conditioning signals.

Table 4: Best depth result for the altitude-interpolation split: training uses all scenes 01–09 at 30 m and 50 m altitude, validation uses scenes 01–04 at 40 m altitude, and testing uses scenes 05–08 at 40 m altitude.

Backbone & Depth head	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
V-JEPA 2.1 RGB FiLM_DPT_RGB	0.9653	0.9997	0.9999	0.1021	4.7071	3.9570	0.119599

Appendix Table ?? shows that the video FiLM-DPT branch is strongest at 30 m, while the RGB FiLM-DPT branch is stronger at 40 m and 50 m. The corresponding scene-wise breakdown is provided in Appendix Table ??: video helps on scene_08, but RGB FiLM-DPT is stronger on scene_09.

4.3 SSC Benchmarking Results

Table 5 summarizes overall SSC results on the OccuFly test split. Detailed altitude-wise numbers and per-class IoU visualizations are provided in Appendix Table ?? and Appendix Figs. ??–??. We abbreviate FoundationSSC as FSSC. The strongest overall semantic result is obtained by the FSSC RGB-context DPT variant with FiLM-DPT depth, while replacing the DAV2 prior with V-JEPA 2.1 FiLM-DPT in the CGFormer-style path improves SSC mIoU with similar SC IoU.

Table 5: Overall OccuFly SSC comparison on the test split. Light beige marks the best SC IoU and light pink marks the best SSC mIoU.

Method	SC IoU $\uparrow$	SSC mIoU $\uparrow$
CGFormer DAV2 prior	36.77	3.05
CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	36.51	3.31
FSSC RGB ctx DPT, depth FiLM-DPT	37.68	3.46
FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	35.47	1.73
FSSC video ctx DTA-DPT, depth DTA-MVS	24.55	0.17
FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	32.55	2.36

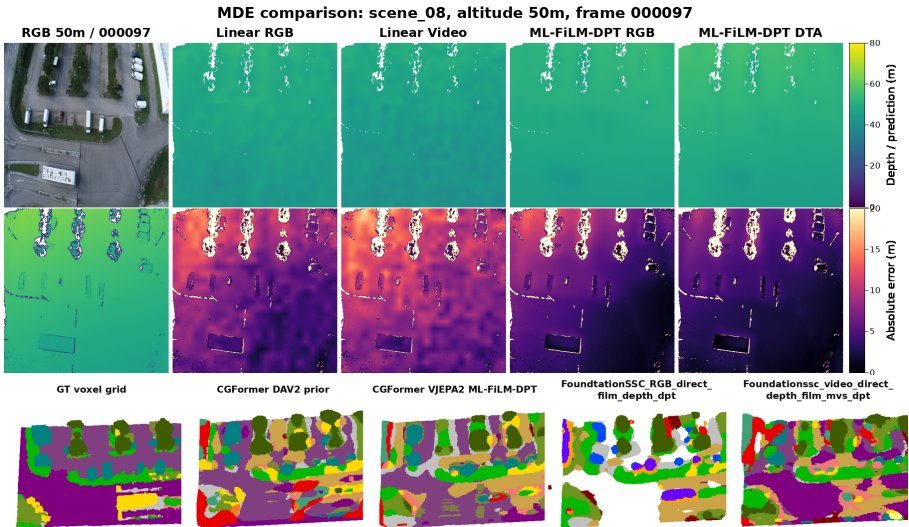


Fig. 6: Qualitative OccuFly depth and SSC example for scene 08 at 50 m altitude, frame 000097.

5 Discussion and Conclusion

5.1 Discussion

The results support three main conclusions. First, frozen V-JEPA 2.1 features contain useful aerial geometry, but metric depth requires dense, scale-aware decoding. In Table 3, the RGB and four-frame video linear probes reach 7.808 m and 7.226 m RMSE, whereas FiLM-DPT RGB achieves the best V-JEPA 2.1 standard-split result with 4.621 m RMSE and 0.129 AbsRel. This gap indicates that frozen transformer features are insufficient without reconstruction capacity and metadata. Because altitude and intrinsics determine the mapping from image evidence to metric rays, the FiLM pathway in Eqs. (13)–(14) is central to aerial depth estimation rather than an auxiliary refinement.

Second, video and altitude cues are helpful but conditional. DTA-FiLM-DPT Video is strongest at 30 m, reaching 2.939 m RMSE in Appendix Table ??, but FiLM-DPT RGB is stronger at 40 m and 50 m, with similar scene dependence in Appendix Table ??. Thus, current V-JEPA 2.1 video tokens provide complementary cues but do not consistently outperform a well-conditioned RGB decoder. The altitude-interpolation result in Table 4 further supports physically meaningful conditioning: after training at 30 m and 50 m, FiLM-DPT RGB reaches $0.9653 \delta_1$, 0.1021 AbsRel, and 4.7071 m RMSE at the held-out 40 m altitude. This result should be interpreted cautiously because scenes 01–08 appear across the interpolation protocol.

Third, improved depth helps SSC only when the depth prior, lifting module, and semantic context remain aligned. In Table 5, replacing the DAV2 prior with the V-JEPA 2.1 FiLM-DPT RGB prior improves SSC mIoU in the CGFormer-style setting while preserving similar SC IoU. The best overall result is the FSSC RGB-context DPT variant with FiLM-DPT depth, reaching 37.68 SC IoU and 3.46 SSC mIoU. By contrast, conditioning both RGB context and depth reduces mIoU to 1.73, suggesting context–geometry mismatch. Pose-aware MVS shows similar sensitivity: FiLM-conditioning the MVS logits improves the unconditioned variant from 24.55 SC IoU / 0.17 SSC mIoU to 32.55 / 2.36, but remains below the best RGB-context configuration. The gap between SC IoU and SSC mIoU also shows that occupancy localization and semantic labeling remain distinct failure modes.

5.2 Limitations and Conclusion

This study isolates frozen-feature transfer and therefore does not test end-to-end finetuning. All depth heads operate from a 24×24 V-JEPA feature grid, video clips lack capture-FPS metadata, and SSC mIoU remains low in absolute terms. Overall, frozen V-JEPA 2.1 features provide useful aerial geometry for OccuFly, but strong aerial 3D perception requires DPT-style dense decoding, altitude/intrinsics conditioning, and lifting architectures whose depth and context pathways remain compatible.

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